

STANDARD WAVEFORMS

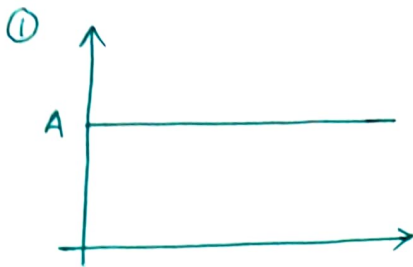
V_{avg}

V_{rms}

Form factor

peak factor

V_{pp}



A

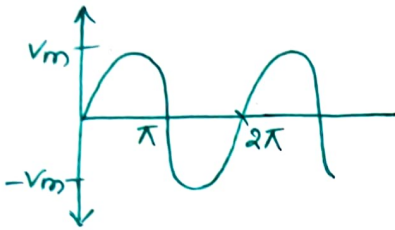
A

1

1

A

(2) $V_m \sin(\omega t + \theta)$



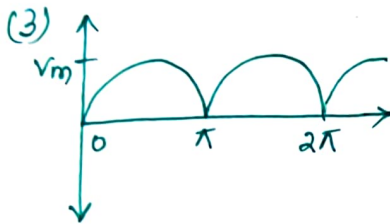
(full cycle)
0
(half cycle)
 $\frac{2V_m}{\pi}$

$\frac{V_m}{\sqrt{2}}$

$\frac{\pi}{2\sqrt{2}} = 1.11$

$\sqrt{2} = 1.41$

$2V_m$



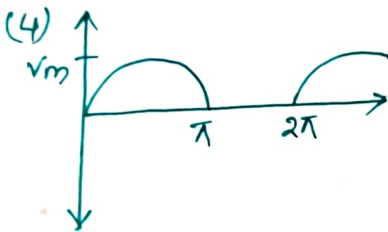
$\frac{2V_m}{\pi}$

$\frac{V_m}{\sqrt{2}}$

1.11

$\sqrt{2}$

V_m



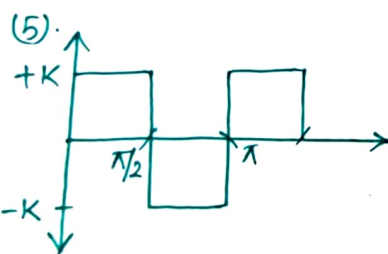
$\frac{I_m}{\pi}$ or $\frac{V_m}{\pi}$

$\frac{V_m}{2}$

1.57

2

V_m



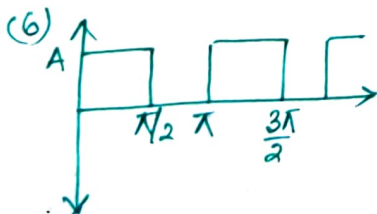
(full cycle)
0
(half cycle)
K

K

1

1

$2K$



$\frac{A}{2}$

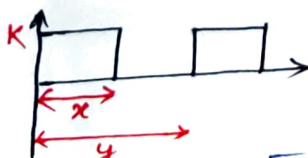
$\frac{A}{\sqrt{2}}$

$\sqrt{2}$

$\sqrt{2}$

A

NOTE:



$AVG = K\left(\frac{x}{y}\right)$ $RMS = K\sqrt{\frac{x}{y}}$

STANDARD WAVEFORM

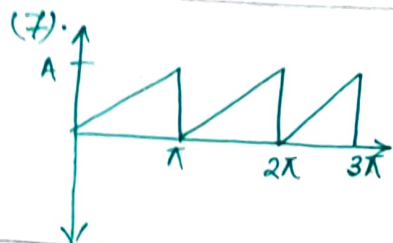
Vavg

Vrms

FF
(RMS/AVG)

PF
(MAX/RMS)

Vpp



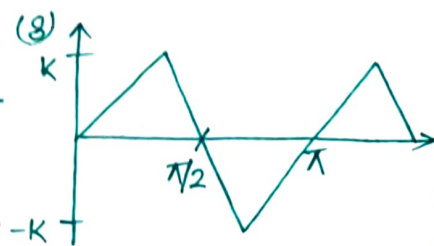
$$\frac{A}{2}$$

$$\frac{A}{\sqrt{3}}$$

$$\frac{2}{\sqrt{3}}$$

$$\sqrt{3}$$

$$A$$



(Half cycle)

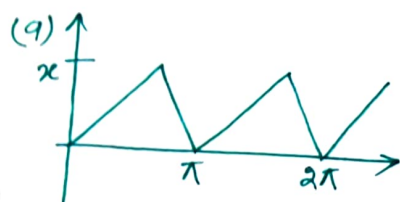
$$\frac{K}{2}$$

$$\frac{K}{\sqrt{3}}$$

$$\frac{2}{\sqrt{3}}$$

$$\sqrt{3}$$

$$2K$$



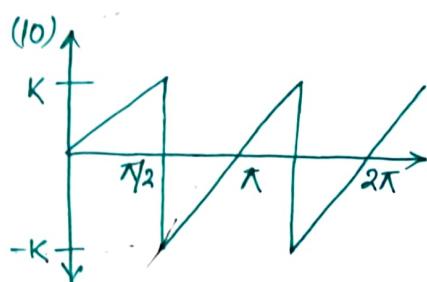
$$\frac{x}{2}$$

$$\frac{x}{\sqrt{3}}$$

$$\frac{2}{\sqrt{3}}$$

$$\sqrt{3}$$

$$x$$



(Full cycle)

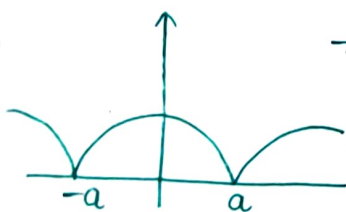
$$0$$

$$\frac{K}{\sqrt{3}}$$

$$\frac{2}{\sqrt{3}}$$

$$\sqrt{3}$$

$$2K$$



$$\rightarrow V_{avg} = \frac{\pi a^2}{2 \times 2a}$$

$$\rightarrow V_{avg} = \frac{\pi a}{4}$$

$$\rightarrow V_{rms} = \sqrt{\frac{2}{3}} a$$

SN	PARAMETER	HALF (RECTIFIER)	FULL	BRIDGE
1	No of diodes	1	2	4
2	PIV	V_m	$2V_m$	V_m
3	I _{RMS}	$I_m/2$	$I_m/\sqrt{2}$	$\frac{I_m}{\sqrt{2}}$
4	I(AVG)	I_m/π	$\frac{2I_m}{\pi}$	$\frac{2I_m}{\pi}$
5	DC O/P V/G	$\frac{V_m}{\pi}$	$\frac{2V_m}{\pi}$	$\frac{2V_m}{\pi}$
6	Ripple factor $RF = \sqrt{FF^2 - 1}$	1.21	0.482	0.482
7	Ripple freq	f	2f	2f
8	FORM FACTOR	1.57	1.11	1.11
9	Rectification efficiency(%)	40.6%	81.2%	81.2%
10	TUF	0.287	0.693	0.812